EXPERIMENT: MONITORING ESTABLISHED TCP CONNECTIONS USING POWERSHELL

Task: T29
Name: Varun Vinod
Roll Number: 30

1. Aim

To use PowerShell (Get-NetTCPConnection and Get-Process) to list established TCP connections with their owning process names, save them as a JSON baseline, alert on new process and remote-endpoint pairs in monitor mode, and compare the result with T27.

2. Course Outcome and Module

Course outcome 3: Analyze live network traffic and identify suspicious activities such as Trojans, backdoors, and sniffers.

Module: Network Connection Monitoring and Analysis

3. Tools and Versions

Tool	Version	Operating System
Windows PowerShell	5.1	Windows 10 or 11
Cmdlets	Get-NetTCPConnection, Get-Process (built in)	Windows 10 or 11

The script uses only built-in cmdlets and was run from the Visual Studio Code terminal. Check the version with $PSVersionTable.PSVersion.

4. Program or Commands

The script net_baseline.ps1 reads all ESTABLISHED TCP connections, maps each owning PID to a process name using Get-Process, and builds a pair of the form process|remote_ip|remote_port. The local port is left out because it changes with every new connection. It was run with:

.\net_baseline.ps1 -Mode Baseline : to save the current pairs as the baseline (baseline_t29.json).
.\net_baseline.ps1 -Mode Monitor : to compare current pairs with the baseline and show the alert.

```
<#
 T29: Established TCP connection baseline and monitor (PowerShell 5.1)
 Usage:  .\net_baseline.ps1 -Mode Baseline
         .\net_baseline.ps1 -Mode Monitor
#>
param(
    [Parameter(Mandatory = $true)]
    [ValidateSet('Baseline', 'Monitor')]
    [string]$Mode,
    [string]$BaselineFile = '.\baseline_t29.json'
)

function Get-EstablishedPairs {
    $names = @{}
    foreach ($p in Get-Process) { $names[[int]$p.Id] = $p.ProcessName }

    Get-NetTCPConnection -State Established -ErrorAction SilentlyContinue |
        ForEach-Object {
            $procName = $names[[int]$_.OwningProcess]
            if (-not $procName) { $procName = 'unknown' }
            [pscustomobject]@{
                Process = $procName
                PID     = $_.OwningProcess
                Local   = "$($_.LocalAddress):$($_.LocalPort)"
                Remote  = "$($_.RemoteAddress):$($_.RemotePort)"
                Pair    = "$procName|$($_.RemoteAddress)|$($_.RemotePort)"
            }
        }
}

$current = @(Get-EstablishedPairs)

if ($Mode -eq 'Baseline') {
    $pairs = @($current | Select-Object -ExpandProperty Pair -Unique)
    ConvertTo-Json -InputObject $pairs | Set-Content -Path $BaselineFile -Encoding UTF8
    Write-Host "[+] Baseline saved to $BaselineFile"
    Write-Host "    $($pairs.Count) unique pairs from $($current.Count) connections"
    return
}

if (-not (Test-Path $BaselineFile)) {
    Write-Host '[!] No baseline found. Run with -Mode Baseline first.' -ForegroundColor Yellow
    return
}

$known = @(Get-Content -Path $BaselineFile -Raw | ConvertFrom-Json)
$new = @($current | Where-Object { $known -notcontains $_.Pair } | Sort-Object Pair -Unique)

Write-Host "[*] $($current.Count) established connections checked"
if ($new.Count -eq 0) {
    Write-Host '[OK] No new process/remote pairs vs baseline.' -ForegroundColor Green
}
else {
    Write-Host ('=' * 70)
    Write-Host " ALERT: $($new.Count) new pair(s) not in baseline" -ForegroundColor Red
    Write-Host ('=' * 70)
    $new | Format-Table Process, PID, Local, Remote -AutoSize
}
```

5. How to Run

Use Windows 10 or 11 with Windows PowerShell 5.1 and save the script as net_baseline.ps1.

Open the folder in the terminal. If scripts are blocked, run Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass for that session only.

Run .\net_baseline.ps1 -Mode Baseline.

Open new pages in the browser, then run .\net_baseline.ps1 -Mode Monitor.

Take screenshots of the baseline and the alert, and prepare the report.

6. Sample Output with Screenshots

6.1 Creation of the Baseline

The -Mode Baseline run found 46 established connections and saved 42 unique pairs to baseline_t29.json.

```
PS C:\Users\ASUS\OneDrive\Desktop> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
PS C:\Users\ASUS\OneDrive\Desktop> .\net_baseline.ps1 -Mode Baseline
[+] Baseline saved to .\baseline_t29.json
 42 unique pairs from 46 connections
PS C:\Users\ASUS\OneDrive\Desktop>
```

Figure T29-1: Baseline created using -Mode Baseline

6.2 Alert for New Pairs

The -Mode Monitor run checked 63 established connections, and 50 new pairs were not in the baseline. They included brave (PID 7044) connecting to external servers, mostly on port 443, and background programs such as RiotClientServices, WUDFHost, ArmouryCrate and asus_framework, many on loopback (127.0.0.1) addresses.

```
PS C:\Users\ASUS\OneDrive\Desktop> .\net_baseline.ps1 -Mode Monitor
[*] 63 established connections checked
================================================================
 ALERT: 50 new pair(s) not in baseline
================================================================

Process                         PID Local            Remote
-------                         --- -----            ------
AcPowerNotification           21548 127.0.0.1:49702  127.0.0.1:1042
ArmouryCrate.Service           6640 127.0.0.1:50100  127.0.0.1:43813
ArmouryCrate.Service           6640 127.0.0.1:50100  127.0.0.1:43818
ArmouryCrate.Service           6640 127.0.0.1:50100  127.0.0.1:43820
ArmouryCrate.UserSessionHelper 46448 127.0.0.1:51100  127.0.0.1:43819
ArmouryCrate.UserSessionHelper 46448 127.0.0.1:51100  127.0.0.1:43821
ArmouryCrate.UserSessionHelper 46448 127.0.0.1:43813  127.0.0.1:50100
ArmouryCrate.UserSessionHelper 46448 127.0.0.1:43824  127.0.0.1:9012
ArmouryCrate                  15580 127.0.0.1:43858  127.0.0.1:13030
ArmouryCrate                  15580 127.0.0.1:43818  127.0.0.1:50100
ArmouryCrate                  15580 127.0.0.1:43819  127.0.0.1:51100
ArmourySocketServer            8164 127.0.0.1:49690  127.0.0.1:1042
ArmourySocketServer            8164 127.0.0.1:9012   127.0.0.1:43824
asus_framework                46392 127.0.0.1:1042   127.0.0.1:49690
asus_framework                46392 127.0.0.1:1042   127.0.0.1:49702
backgroundTaskHost              656 127.0.0.1:43820  127.0.0.1:50100
backgroundTaskHost              656 127.0.0.1:43821  127.0.0.1:51100
brave                          7044 192.168.1.2:32472 142.251.223.14:443
brave                          7044 192.168.1.2:17114 149.154.167.99:443
brave                          7044 192.168.1.2:5213  149.154.170.200:443
brave                          7044 192.168.1.2:35371 149.154.174.200:443
brave                          7044 192.168.1.2:2275  151.101.194.137:443
brave                          7044 192.168.1.2:22400 172.235.25.160:443
brave                          7044 192.168.1.2:17416 3.6.199.65:443
brave                          7044 192.168.1.2:16663 74.125.130.132:443
DSAService                     5280 192.168.1.2:63150 103.94.139.202:443
FileCoAuth                    17876 192.168.1.2:63148 20.42.65.95:443
ipfsvc                         5360 127.0.0.1:49731  127.0.0.1:49730
ipfsvc                         5360 127.0.0.1:49730  127.0.0.1:49731
msedgewebview2                33028 192.168.1.2:39622 57.144.57.32:5222
NVDisplay.Container            3516 127.0.0.1:49721  127.0.0.1:49720
NVDisplay.Container            3516 127.0.0.1:49720  127.0.0.1:49721
oracle                         6504 ::1:24058        ::1:1521
QHActiveDefense                4092 192.168.1.2:3544  54.169.7.73:80
QHSafeTray                    43820 192.168.1.2:1980  54.255.213.32:80
Riot Client                   43104 127.0.0.1:58142  127.0.0.1:41819
RiotClientServices            25584 192.168.1.2:42465 104.18.40.25:443
RiotClientServices            25584 127.0.0.1:41819  127.0.0.1:58142
RiotClientServices            25584 127.0.0.1:41819  127.0.0.1:58183
RiotClientServices            25584 127.0.0.1:41819  127.0.0.1:9271
RiotClientServices            25584 192.168.1.2:63156 172.64.153.253:443
RiotClientServices            25584 192.168.1.2:24481 172.65.217.212:5223
ROGLiveService                 5540 127.0.0.1:13030  127.0.0.1:43858
SearchHost                    23608 192.168.1.2:63080 52.98.58.50:443
svchost                        5740 192.168.1.2:49453 4.213.25.242:443
tnslsnr                        6072 ::1:1521         ::1:24058
WUDFHost                       1816 127.0.0.1:49717  127.0.0.1:49716
WUDFHost                       1816 127.0.0.1:49716  127.0.0.1:49717
WUDFHost                      12156 127.0.0.1:49719  127.0.0.1:49718
WUDFHost                      12156 127.0.0.1:49718  127.0.0.1:49719
```

Figure T29-2: Alert showing new pairs not in the baseline

Loopback pairs add extra alerts because the remote port of a local connection is the other program's port, which changes between runs.

7. Result

Established TCP connections were read with Get-NetTCPConnection, matched to process names with Get-Process, and saved as a JSON baseline of 42 pairs from 46 connections. The monitor run checked 63 connections and reported 50 new pairs, including brave (PID 7044) and several background programs. Only Windows was used in this experiment.

8. Comparison with T27

T27 used Python with psutil for the same kind of baseline check. T29 does it in PowerShell with built-in cmdlets.

Feature	T27 (Python, psutil)	T29 (PowerShell)
Install needed	Python and the psutil package	None, built into Windows
Connection source	psutil.net_connections()	Get-NetTCPConnection -State Established
Process name	psutil.Process(pid).name()	Get-Process, matched by PID
Protocols	TCP and UDP possible	TCP only (UDP needs Get-NetUDPEndpoint)
Platforms	Windows, Linux, macOS	Windows only (PowerShell 5.1)
Baseline format	As written in T27	JSON list of process|remote_ip|remote_port
Alert	As shown in T27	Red ALERT banner and table of new pairs

Both detect a new connection by comparing against a saved baseline. T29 is simpler to deploy on Windows, while T27 is portable across operating systems.

9. Viva Questions and Answers

1. What does Get-NetTCPConnection return?

It returns TCP connections with local and remote address and port, state, and the OwningProcess PID.

2. Why is Get-Process used?

Get-NetTCPConnection gives only the PID, and Get-Process gives the process name for that PID.

3. What is a baseline?

A saved list of normal connections. A pair not in the list is reported as an alert.

4. Why is the local port left out of the pair?

The local port changes for every new connection, so including it would raise false alerts.

5. What is the difference between this script and netstat -ano?

netstat gives text that must be parsed, while PowerShell returns objects that can be filtered and saved directly as JSON.

10. Common Errors: Message, Cause, and Fix

Error or Problem	Possible Cause	Solution
Script cannot be loaded because running scripts is disabled	The execution policy blocks unsigned scripts.	Run Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass.
No baseline found. Run with -Mode Baseline first.	Monitor mode was run before a baseline was saved.	Run the script with -Mode Baseline first.
Process name shows unknown	The process ended before Get-Process ran, or its PID was reused.	Run again; connections are short-lived.
Many alerts at once	Browsers and background programs open many connections to new servers, and loopback ports change.	Re-learn the baseline after normal use, or ignore loopback addresses in the script.

Declaration: AI tools were utilized for assistance in developing and debugging the program, preparing the report, and preparing, reviewing, and verifying viva answers.